

Polynomial identities, sums of three squares, and quaternionic (anti-)automorphisms

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Abstract

Evaluating words in a free monoid at the quaternions $t + iu$ and $x + yj + zk$, we show that every word fixed by the composition of string reversal with inversion of the α -letters evaluates to a quaternion with vanishing i -coefficient, and hence yields a polynomial identity

$$(t^2 + u^2)^{2m} (x^2 + y^2 + z^2)^n = f_1^2 + f_j^2 + f_k^2$$

with $f_1, f_j, f_k \in \mathbb{Z}[t, u, x, y, z]$. The proof is a two-line fixed-point argument for the anti-involution $p \mapsto i^{-1}\bar{p}i$ and is valid over every commutative ring. Using the freeness of a rotation group of Świerczkowski we prove that the evaluation map is injective on reduced words, and we determine the exact number of identities of each bidegree: the count has rational generating function $(1 + s)/(1 - s - q - 3sq)$, so it grows linearly in n for $m = 1$ (namely $2n$ or $2n - 2$ identities), polynomially of degree m in n for each fixed m , and exponentially, with growth rate 9, along diagonals. We situate the family against the classical composition problem (no bilinear identity of this shape exists, and by a theorem of Pfister the exponent $2m$ cannot be replaced by any odd number when n is odd, even allowing rational functions), identify the underlying mechanism as a word-indexed twisting of the conjugation action of $\mathbb{H}^\times$ on pure quaternions, classify the anti-involutions that can play the role of $p \mapsto i^{-1}\bar{p}i$, and settle the group-theoretic side of the remaining uniqueness question by a freeness theorem for the rotations through $\arccos \frac{1}{3}$ about the three coordinate axes, proved by a rank-one-modulo-3 refinement of Świerczkowski's method; what remains is a single symmetric-word condition, verified through length twelve.

1 Introduction

A classical identity recorded in Dickson's chapter on sums of three squares [1, Ch. VII, p. 265] and attributed there to V.-A. Lebesgue (1868) expresses the square of a sum of four squares

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as a sum of three:

$$(\alpha^2 + \beta^2 + \gamma^2 + \delta^2)^2 = (\alpha^2 + \beta^2 - \gamma^2 - \delta^2)^2 + (2\alpha\gamma + 2\beta\delta)^2 + (2\alpha\delta - 2\beta\gamma)^2 : \quad (1)$$

in quaternionic terms, conjugating i by a quaternion q produces a pure quaternion of norm $N(q)^2$. The present paper studies a cognate family in which the conjugating quaternion is itself structured—a word in prescribed letters—beginning with

$$(t^2 + u^2)^2 (x^2 + y^2 + z^2) = ((t^2 + u^2)x)^2 + ((t^2 - u^2)y - 2tuz)^2 + ((t^2 - u^2)z + 2tuy)^2, \quad (2)$$

which expresses a product of quadratic forms in *independent* variables as a sum of three polynomial squares. (Identity (2), though elementary, does not appear in Dickson's chapter, and we have not located it elsewhere.) The identity is not bilinear: the first factor enters through the *square* of $t^2 + u^2$, and the polynomials on the right are not bilinear in the two groups of variables; neither feature can be repaired. By the Hurwitz–Radon theory of compositions of quadratic forms [2, 3, 4, 9, 10], no identity of size $[2, 3, 3]$ or $[3, 3, 3]$ exists over any field of characteristic $\neq 2$ —consistently, the classical product identities of the shape $(a^2 + b^2 + c^2)(p^2 + q^2 + r^2) = \square + \square + \square$ recorded by Dickson (Weddle, Young, Catalan; [1, Ch. VII, pp. 263, 269]) all hold only under constraints binding the two sets of variables—and, much more strongly, a theorem of Pfister [6] implies that $(t^2 + u^2)^a (x^2 + y^2 + z^2)^n$ with a and n odd is not a sum of three squares even in $\mathbb{R}(t, u, x, y, z)$ (Proposition 4.2). The doubling of the exponent in (2) is forced.

This paper studies the natural habitat of (2): a family of identities

$$(t^2 + u^2)^{2m} (x^2 + y^2 + z^2)^n = f_1^2 + f_j^2 + f_k^2, \quad f_1, f_j, f_k \in \mathbb{Z}[t, u, x, y, z], \quad (3)$$

indexed by words in a free monoid. Let h be the evaluation sending the letters $\alpha, \alpha^{-1}, \beta, \beta^{-1}$ to the quaternions

$$h(\alpha) = t + iu, \quad h(\alpha^{-1}) = t - iu, \quad h(\beta) = x + yj + zk, \quad h(\beta^{-1}) = x - yj - zk,$$

and extend multiplicatively to words. Let S denote the set of reduced words fixed by the composition of string reversal with inversion of all α -letters. The basic phenomenon (Theorem 3.2) is that for $\sigma \in S$ the coefficient of i in $h(\sigma)$ vanishes identically, so that taking norms produces an identity (3), where $2m$ and n count the α -letters and β -letters of σ . The word $\alpha\beta\alpha^{-1}$ recovers (2).

Individually, these identities are classical in mechanism if not in statement: we show in §5 that each is a word-indexed twisting of the conjugation action of $\mathbb{H}^\times$ on pure quaternions, whose use in the study of ternary forms goes back at least to Pall [5]; the $m = 0$ subfamily is the folklore family of identities for $(x^2 + y^2 + z^2)^n$, governed by Chebyshev polynomials. What we believe is new is the *combinatorial parametrization and its exactness*: the correspondence between identities and symmetric words is faithful. Our main results are:

- (i) **Cancellation** (Theorem 3.2). For $\sigma \in S$, $h(\sigma)$ has no i -component. The proof exhibits $h(\sigma)$ as a fixed point of the anti-involution $\theta(p) = i^{-1}\bar{p}i$; the theorem holds over every commutative ring, by specialization from $\mathbb{Z}$ or by a companion induction that avoids division by 2 entirely. Among the involutions of the quaternions, θ is, up to symmetry, the only one that could produce a nontrivial family of this kind (Proposition 5.1).

- (ii) **Injectivity** (Theorem 6.1). The evaluation h is injective on reduced words, and $h(\sigma) = \pm h(\sigma')$ forces $\sigma = \sigma'$. The proof passes to the rotation group $\mathrm{SO}(3)$ and invokes the freeness, due to Świerczkowski [11], of the group generated by rotations through $\arccos \frac{1}{3}$ about two perpendicular axes.
- (iii) **Census** (Theorem 6.3). Up to the sign changes that obviously preserve an identity, the identities of type (m, n) correspond to the orbits of an explicit involution on S , and their number $c(m, n)$ satisfies

$$\sum_{m, k \geq 0} c(m, 2k+1) s^m q^k = \frac{1+s}{1-s-q-3sq}, \quad c(m, 2k+2) = c(m, 2k+1).$$

In particular $c(1, n)$ equals $2n$ for n odd and $2n-2$ for n even; for each fixed m the count grows as a polynomial of degree m in n ; and along the diagonal $c(k, 2k+1)^{1/k} \rightarrow 9$. The growth is not exponential in $m+n$: rows grow polynomially; only diagonals grow exponentially.

- (iv) **Fine equivalence** (§6.3). Allowing also permutations of the three squares, a parity invariant shows only the transposition of f_j with f_k is in question, and a degeneration argument at $u = 0$ confines any further coincidence to a narrow class of word pairs. A freeness theorem for the rotations through $\arccos \frac{1}{3}$ about the three coordinate axes (Theorem 6.6), proved by a rank-one-modulo-3 refinement of Świerczkowski's method, then excludes every sign-pattern coincidence outright; the residue is a single symmetric-word condition (Conjecture 6.9), verified through word length 12.

We work throughout in a complex compression of the quaternions ($\mathbb{H} = \mathbb{C} \oplus \mathbb{C}j$, $\zeta = t + iu$, $w = y + iz$) under which each identity (3) takes the form

$$(\zeta \bar{\zeta})^{2m} (x^2 + w\bar{w})^n = f_1^2 + F\bar{F}, \quad F = f_j + if_k,$$

and (2) becomes the one-line statement $f_1 = \zeta \bar{\zeta} x$, $F = \zeta^2 w$. The appendix tabulates, in this notation, every identity through word length 6 and a sample beyond; all tables, and every identity displayed in the paper, have been generated and verified by machine (Remark 7.4).

For orientation within the literature on sums of squares of polynomials: the counting of essentially different representations has precedent—a smooth complex ternary quartic has exactly 63 inequivalent representations as a sum of three squares [8], exactly 8 of them real when the quartic is positive semidefinite with smooth zero curve [7]—but there the count attaches to a single form, while here it attaches to a bihomogeneous family, and the enumeration is by words rather than by Gram matrices. We have not found the present family, or its census, in the literature; in particular Shapiro's monograph [9], which surveys the composition problem comprehensively, contains nothing of this bihomogeneous word-indexed shape—its treatment of nonbilinear phenomena (Hopf constructions, quadratic maps between spheres) concerns a different question.

2 Words, evaluations, and complex coordinates

2.1 Words

Let W be the free monoid on the four letters $\alpha, \alpha^{-1}, \beta, \beta^{-1}$. A word is *reduced* if no letter is adjacent to its formal inverse. Reduced words are exactly the normal forms for elements of the free group $F_2 = \langle \alpha, \beta \rangle$; we will pass between the two points of view freely, but we emphasize that our evaluation h below is defined on W , *not* on F_2 : since $h(\alpha)h(\alpha^{-1}) = t^2 + u^2 \neq 1$, the map h does not respect the group relations, and statements must be phrased for words.

Three letterwise involutions act on W : ι_α inverts each $\alpha^{\pm 1}$, ι_β inverts each $\beta^{\pm 1}$, and their composite inverts every letter. Let ρ reverse strings. All of these preserve reducedness, commute with one another, and ρ commutes with the letterwise maps.

Definition 2.1. S is the set of reduced words σ with $\rho\iota_\alpha(\sigma) = \sigma$: reading σ backwards while inverting the α -letters returns σ .

The set S is preserved by ι_α and by ι_β (for ι_β : $\rho\iota_\alpha$ commutes with ι_β ; for ι_α : the condition $\rho\iota_\alpha(\iota_\alpha\sigma) = \iota_\alpha\sigma$ unwinds to $\rho\sigma = \iota_\alpha\sigma$, which is a restatement of $\sigma \in S$). Moreover, for $\sigma \in S$ the reduced word representing σ^{-1} in F_2 , namely ρ applied to the letterwise inverse of σ , equals $\iota_\beta(\sigma)$. Inversion of symmetric words is thus already accounted for by ι_β .

Lemma 2.2 (Bisection). *Every $\sigma \in S$ factors uniquely as*

$$\sigma = v \cdot \beta^\varepsilon \cdot \iota_\alpha \rho(v), \quad \varepsilon \in \{1, -1, 2, -2\},$$

with v a reduced word (possibly empty) not ending in $\beta^{-\text{sgn } \varepsilon}$. Conversely every such pair (v, ε) produces an element of S . In particular the number of α -letters of any $\sigma \in S$ is even.

Proof. Write ℓ for the length of σ and number its letters $1, \dots, \ell$; the defining symmetry says letter $\ell + 1 - r$ is ι_α (letter r). If ℓ is odd the middle letter c satisfies $c = \iota_\alpha(c)$, so $c \in \{\beta, \beta^{-1}\}$ and $\varepsilon = \pm 1$. If ℓ is even the middle two letters are $c, \iota_\alpha(c)$; reducedness rules out $c \in \{\alpha, \alpha^{-1}\}$ (which would give $\alpha\alpha^{-1}$ or $\alpha^{-1}\alpha$), leaving $\beta\beta$ or $\beta^{-1}\beta^{-1}$, so $\varepsilon = \pm 2$. The prefix v preceding the middle block is reduced, and the junction conditions at both ends of the block amount to the single condition stated (the right-hand junction asks that ι_α of the last letter of v not cancel $\beta^{\text{sgn } \varepsilon}$, which is the same condition, since ι_α fixes β -letters and moves α -letters to α -letters). The converse and the uniqueness are immediate. The α -letters of σ pair off under the symmetry, none being central; hence their number is even. $\square$

We write $2m$ for the number of α -letters and n for the number of β -letters of $\sigma \in S$, and call (m, n) the *type* of σ .

2.2 Evaluations, real and complex

Let R be a commutative ring and $\mathbb{H}_A = A \oplus Ai \oplus Aj \oplus Ak$ the quaternion algebra over a commutative R -algebra A ; $\bar{p}$ denotes the standard involution and $N(p) = p\bar{p}$ the norm. Over $A = R[t, u, x, y, z]$ define the multiplicative map $h: W \rightarrow \mathbb{H}_A$ on letters by

$$h(\alpha^{\pm 1}) = t \pm iu, \quad h(\beta^{\pm 1}) = x \pm (yj + zk).$$

Since each letter has norm $t^2 + u^2$ or $x^2 + y^2 + z^2$ and N is multiplicative,

$$N(h(\sigma)) = (t^2 + u^2)^{2m}(x^2 + y^2 + z^2)^n \quad \text{for } \sigma \text{ of type } (m, n). \quad (4)$$

It clarifies both proofs and formulas to split off the subfield $\mathbb{C} = R[i]$: write $\mathbb{H} = \mathbb{C} \oplus \mathbb{C}j$, so that a quaternion is a pair (z_1, z_2) of “complex” entries, with multiplication

$$(z_1 + z_2j)(z_3 + z_4j) = (z_1z_3 - z_2\bar{z}_4) + (z_1z_4 + z_2\bar{z}_3)j. \quad (5)$$

Introduce

$$\zeta = t + iu, \quad w = y + iz,$$

so that $h(\alpha^{\pm 1}) = \zeta^{\pm}$ (meaning ζ or $\bar{\zeta}$) and $h(\beta^{\pm 1}) = x \pm wj$. Every $h(\sigma)$ is then computed inside $\mathbb{C}[\zeta, \bar{\zeta}, x, w, \bar{w}]$ by iterating (5). A quaternion has vanishing i -coefficient exactly when its first complex entry z_1 is fixed by the conjugation $\zeta \leftrightarrow \bar{\zeta}$, $w \leftrightarrow \bar{w}$; in that case, writing $z_1 = f_1$ and $z_2 = F = f_j + if_k$, the norm is

$$N(z_1 + z_2j) = f_1^2 + F\bar{F} = f_1^2 + f_j^2 + f_k^2. \quad (6)$$

3 The cancellation theorem

Define $\theta: \mathbb{H}_A \rightarrow \mathbb{H}_A$ by

$$\theta(p) = i^{-1} \bar{p} i.$$

Lemma 3.1. *θ is an A -linear anti-involution with*

$$\theta(1, i, j, k) = (1, -i, j, k),$$

so that $\theta(p) = p$ if and only if twice the i -coefficient of p vanishes; in particular $\text{Fix}(\theta) = A \oplus Aj \oplus Ak$ whenever 2 is a non-zero-divisor in A . Moreover, for every word $\tau \in W$,

$$\theta(h(\tau)) = h(\rho \iota_\alpha(\tau)).$$

Proof. θ is a composite of the anti-automorphism $p \mapsto \bar{p}$ with the inner automorphism $p \mapsto i^{-1}pi$, hence an anti-automorphism; its square is conjugation by $\bar{i}i = N(i)$, the identity. The values on $1, i, j, k$ are immediate ($\theta(j) = i^{-1}(-j)i = j$, and likewise for k), giving the fixed module. For the intertwining: θ reverses products, fixes $h(\beta^{\pm 1}) \in A \oplus Aj \oplus Ak$, and swaps $h(\alpha) = t + iu$ with $h(\alpha^{-1}) = t - iu$; so applying θ to $h(\tau) = h(\tau_1) \cdots h(\tau_\ell)$ yields $h(\iota_\alpha \tau_\ell) \cdots h(\iota_\alpha \tau_1) = h(\rho \iota_\alpha \tau)$. $\square$

Theorem 3.2 (Cancellation). *Let R be any commutative ring and $\sigma \in S$. Then $h(\sigma) \in A \oplus Aj \oplus Ak$: the coefficient of i in $h(\sigma)$ vanishes identically.*

Proof. $\theta(h(\sigma)) = h(\rho \iota_\alpha \sigma) = h(\sigma)$, so twice the i -coefficient of $h(\sigma)$ vanishes. Over $R = \mathbb{Z}$ the coefficient, an integer polynomial, therefore vanishes itself; the case of arbitrary R follows by applying to this integer identity the ring homomorphism determined by the parameters t, u, x, y, z . $\square$

Second proof, without reference to 2. A word fixed by $\rho\iota_\alpha$ is empty, a single β -letter, or of the form $c\sigma'\iota_\alpha(c)$ with σ' again fixed (the first and last letters correspond under the symmetry, as do the interiors). By induction it suffices to check that for each letter c the map $p \mapsto h(c)p h(\iota_\alpha c)$ preserves the vanishing of the i -coefficient. For $c = \alpha^{\pm 1}$: $(t \pm iu)(p_0 + p_2j + p_3k)(t \mp iu) = (t^2 + u^2)p_0 + (t \pm iu)^2(p_2j + p_3k)$, and multiplying $p_2j + p_3k$ by a complex number produces no 1- or i -component. For $c = \beta^{\pm 1}$, where $\iota_\alpha(c) = c$: writing $v = h(c) = v_0 + v_2j + v_3k$ and $p = p_0 + p_2j + p_3k$, the i -coefficient of $v p v$ is

$$(v_2p_3 - v_3p_2)v_0 + (v_0p_2 + v_2p_0)v_3 - (v_0p_3 + v_3p_0)v_2 = 0.$$

These are polynomial identities with integer coefficients, so the argument is valid over every commutative ring directly; it is also the form of the theorem best suited to formal verification. $\square$

Remark 3.3. One can argue instead by comparing the automorphism $\mu: (i, j, k) \mapsto (-i, k, j)$ with the anti-automorphism $\nu: (i, j, k) \mapsto (i, k, j)$: each relates $h(\sigma)$ to the same transform of itself, and comparing the two conclusions yields $f_i = -f_i$. Since $\mu\nu = \theta$, this is the same symmetry as in the first proof, differently packaged; like the first proof it establishes $2f_i = 0$ and closes through the universal case $R = \mathbb{Z}$. Only the second proof dispenses with 2 altogether. In every formulation (3) is an identity of integer polynomials and specializes to every commutative ring, including those of characteristic 2.

Remark 3.4. In complex coordinates the theorem reads: for $\sigma \in S$ the first entry of the pair $h(\sigma)$, computed by iterating (5) from the letter values $\zeta^\pm$ and $x \pm wj$, is invariant under $\zeta \leftrightarrow \bar{\zeta}$, $w \leftrightarrow \bar{w}$. This is the form in which the appendix tables should be read, and the form in which machine verification is performed.

Combining Theorem 3.2 with (4) and (6):

Corollary 3.5. *Each $\sigma \in S$ of type (m, n) yields an identity*

$$I(\sigma) : \quad (t^2 + u^2)^{2m}(x^2 + y^2 + z^2)^n = f_1^2 + f_j^2 + f_k^2$$

with $f_1, f_j, f_k \in \mathbb{Z}[t, u, x, y, z]$ bihomogeneous of bidegree $(2m, n)$ in (t, u) and (x, y, z) .

Example 3.6. $\sigma = \alpha\beta\alpha^{-1}$: by (5), $h(\sigma) = (\zeta, 0)(x, w)(\bar{\zeta}, 0) = (\zeta\bar{\zeta}x, \zeta^2w)$, recovering (2) in the form $(\zeta\bar{\zeta})^2(x^2 + w\bar{w}) = (\zeta\bar{\zeta}x)^2 + (\zeta^2w)(\bar{\zeta}^2\bar{w})$. More generally $\alpha^m\beta^n\alpha^{-m}$ gives $f_1 = (\zeta\bar{\zeta})^m P_n$, $F = \zeta^{2m} Q_n w$, where P_n, Q_n are the polynomials of the classical family for $(x^2 + y^2 + z^2)^n$ described in §5. The words that interleave α 's and β 's—the first appearing at type $(1, 3)$ —leave this classical orbit; see the appendix.

4 Structure of the identities

Proposition 4.1 (Parity). *For $\sigma \in S$, the polynomial f_1 is invariant, and f_j, f_k are anti-invariant, under $(y, z) \mapsto (-y, -z)$. Thus every monomial of f_1 has even, and every monomial of f_j, f_k odd, total degree in (y, z) .*

Proof. The substitution $(y, z) \mapsto (-y, -z)$ carries $h(\beta^{\pm 1})$ to $h(\beta^{\mp 1})$ and fixes $h(\alpha^{\pm 1})$; hence it carries $h(\sigma)$ to $h(\iota_\beta\sigma)$. On the other hand $h(\iota_\beta\sigma) = i h(\sigma) i^{-1}$: conjugation by i fixes

$h(\alpha^{\pm 1})$ and carries $h(\beta^{\pm 1})$ to $h(\beta^{\mp 1})$, and two multiplicative maps agreeing on letters agree on words. Since conjugation by i fixes 1 and negates j, k , we conclude $f_1 \mapsto f_1$, $f_{j,k} \mapsto -f_{j,k}$ under the substitution. $\square$

Proposition 4.2 (Parity classification of the binary exponent). *Let $a \geq 1$ be odd and $n \geq 1$, and write $c = t^2 + u^2$, $s = x^2 + y^2 + z^2$.*

- (i) *If n is even, then $c^a s^n = (t c^{(a-1)/2} s^{n/2})^2 + (u c^{(a-1)/2} s^{n/2})^2$ is a (degenerate) sum of two polynomial squares.*
- (ii) *If n is odd, then $c^a s^n$ is not a sum of three squares in $\mathbb{R}(t, u, x, y, z)$, let alone in $\mathbb{R}[t, u, x, y, z]$.*

Thus the doubling of the exponent on $t^2 + u^2$ in (3) is forced exactly when three squares are genuinely needed.

Proof. (i) is displayed. For (ii), a three-square expression for $c^a s^n$ in $\mathbb{R}(t, u, x, y, z)$ yields one for cs after division by the square of $c^{(a-1)/2} s^{(n-1)/2}$. By a theorem of Pfister [6, Satz 3 and its Folgerung], in the form given by Shapiro [9, Prop. 14.6 and Thm. 14.8]: over a field F for which the n -dimensional unit form is anisotropic (such as $F = \mathbb{R}$ with $n = 3$), a rational-function formula $(x_1^2 + x_2^2)(y_1^2 + y_2^2 + y_3^2) = z_1^2 + z_2^2 + z_3^2$ exists if and only if $2 \circ 3 \leq 3$, where $\circ$ is the Stiefel–Hopf–Pfister function. But $2 \circ 3 = 4$, since $\binom{3}{2}$ is odd. $\square$

Remark 4.3. For the basic case $(a, n) = (1, 1)$ there is an elementary alternative: decomposing hypothetical f_i with $\sum f_i^2 = cs$ into bihomogeneous components, the lexicographically extreme components must have bidegree $(1, 1)$ (extreme squares cannot cancel in a sum of real squares), so the f_i would be bilinear, say $(f_1, f_2, f_3) = (tA_1 + uA_2)\mathbf{x}$ for real 3×3 matrices A_1, A_2 . Comparing coefficients of t^2 , u^2 , and tu forces A_1, A_2 orthogonal with $B := A_1^\top A_2$ skew-symmetric; but a 3×3 matrix cannot be both orthogonal and skew-symmetric, its determinant being at once ± 1 and—from $\det B = \det B^\top = \det(-B) = -\det B$ —zero. (This is the first case of the Hurwitz–Radon theorem, $\rho(3) = 1$ [9, Ch. 0, pp. 3–4].) Over $\mathbb{Q}$ the impossibility of a bilinear three-square identity goes back to Legendre (1830), via $3 \cdot 21 = 63$ (see [9, Ch. 0]).

Remark 4.4. For n odd, none of f_1, f_j, f_k can vanish identically in any identity of shape (3): if one vanished, $(x^2 + y^2 + z^2)^n$ would become, upon specializing $(t, u) = (1, 0)$, a sum of two squares $A^2 + B^2$ in $\mathbb{R}[x, y, z]$; factoring $A^2 + B^2 = (A + iB)(A - iB)$ in the unique factorization domain $\mathbb{C}[x, y, z]$, where $x^2 + y^2 + z^2$ is irreducible, forces $A - iB = \overline{A + iB}$ to be an associate of the same power of $x^2 + y^2 + z^2$ as $A + iB$, making n even. (This is the mechanism of Wood’s theorem on polynomial maps between spheres; see [9, Ch. 15, Appendix].) Computationally, no f vanishes in any identity through length 12, for either parity of n .

5 Classical transport, equivariance, and the choice of θ

5.1 Transport to pure-quaternion conjugation

The bisection lemma exhibits $h(\sigma) = qv\theta(q)$ with $q = h(v)$ and $v = h(\beta^\varepsilon)$. Right multiplication by i^{-1} intertwines θ with the standard involution: $\theta(p)i^{-1} = i^{-1}\bar{p}$, so

$$h(\sigma)i^{-1} = q(vi^{-1})\bar{q}.$$

Since $v \in \mathbb{R} \oplus \mathbb{R}j \oplus \mathbb{R}k$, the element vi^{-1} is a *pure* quaternion, and $p \mapsto qp\bar{q}$ is $N(q)$ times the rotation by which the class of q acts on pure quaternions. Thus every identity in the family is the classical mechanism—the double cover $\mathbb{H}^\times \rightarrow \mathrm{SO}(3)$ acting on the pure part—applied with a *word-valued* rotation $q = h(v)$, and transported from the pure coordinate frame (i, j, k) to the frame $(1, j, k)$ by the unit i . The arithmetic exploitation of the mechanism is likewise classical, and sharper than mere precedent: Pall [5, §8] proves that *every* representation of hm^2 as a sum of three integer squares satisfying a mild local condition arises as $y = tx\bar{t}$ from a pure x of norm h and a t of norm m ; the integer specializations of the present family are thus not only anticipated but exhausted by the classical theory. The number-theoretic content of any single identity is therefore classical; the content of the family lies in which rotations arise, and Theorem 6.1 will show the words are faithfully recorded.

5.2 Torus equivariance

Let $c = e^{i\phi/2}$ be a unit complex number. Conjugation by c fixes $\mathbb{C}$ pointwise and, by (5), sends $x + wj$ to $x + c^2w\bar{j}$. Applying this to a word gives

$$ch(\sigma)(\zeta, x, w)c^{-1} = h(\sigma)(\zeta, x, c^2w),$$

whence f_1 is invariant under the phase rotation $w \mapsto e^{i\phi}w$ while F transforms with weight one:

$$F(\zeta, x, e^{i\phi}w) = e^{i\phi}F(\zeta, x, w).$$

Equivalently, every monomial of F has w -degree minus $\bar{w}$ -degree equal to 1—visible throughout the appendix. Each identity is thus equivariant for the circle acting by rotation in the (y, z) -plane and on the pair (f_j, f_k) ; the triples (f_1, f_j, f_k) are matrix-coefficient combinations for this action. (No analogous statement holds for a circle acting on (t, u) : the letters α, α^{-1} evaluate to ζ and $\bar{\zeta}$, which a phase rotation does not permute; and indeed the appendix rows mix powers of ζ and $\bar{\zeta}$.)

5.3 The subfamily $m = 0$ and Chebyshev polynomials

For $\sigma = \beta^n$, the commutative subring $\mathbb{R}[x, \omega]$ with $\omega = wj$, $\omega^2 = -w\bar{w}$, gives

$$\begin{aligned} (x + wj)^n &= P_n(x, w\bar{w}) + Q_n(x, w\bar{w})wj, \\ P_n + Q_n\tau &= \text{the expansion of } (x + \tau)^n \text{ at } \tau^2 = -w\bar{w}, \end{aligned}$$

so that $P_n(x, r^2) = \mathrm{Re}(x + ir)^n$ and $rQ_n(x, r^2) = \mathrm{Im}(x + ir)^n$: the classical (Chebyshev-governed) identities for $(x^2 + y^2 + z^2)^n$. That every power of a sum of three squares is again a sum of three squares is a theorem of Catalan, with explicit identities for small powers—his cube identity (see [1, Ch. VII, pp. 267–268]) is the type-(0, 3) row of our family up to relabeling; the closed form via Re and Im appears to be folklore. The word construction embeds this family (as the α -free words) and deforms it (as soon as α 's interleave).

5.4 Which coordinate can be killed?

The proof of Theorem 3.2 used only two properties of θ : it is an anti-involution, and it interchanges $h(\alpha)$ with $h(\alpha^{-1})$ while fixing the $h(\beta^{\pm 1})$. It is natural to ask what other

involutions could serve, i.e., whether coordinates other than i can be cancelled by symmetric words.

Proposition 5.1. *Over a field of characteristic 0 (or $\mathbb{R}$), every anti-automorphism of $\mathbb{H}$ is the composite of the standard involution with an inner automorphism, and the involutive ones fall into exactly two classes:*

- (a) *the standard involution $p \mapsto \bar{p}$, with fixed space the scalars;*
- (b) *the maps $p \mapsto v \bar{p} v^{-1}$ with v a nonzero pure quaternion, with fixed space the span of 1 and the plane of pure quaternions orthogonal to v .*

All involutions of class (b) are conjugate under inner automorphisms; θ is the case $v = i$. The analogue of the construction for class (a)—words σ equal to their formal inverse-reversal, forcing $h(\sigma)$ scalar—produces only the trivial identities $N(h(v))^2 \cdot (\text{norm of middle letter})$ obtained by doubling. Hence, up to symmetry, the three-dimensional cancellation of §3 is the unique nontrivial instance of the mechanism.

Proof. By Skolem–Noether every automorphism of $\mathbb{H}$ is inner, and every anti-automorphism is $p \mapsto q \bar{p} q^{-1}$ for some $q \in \mathbb{H}^\times$. Such a map is involutive iff $q \bar{q}^{-1}$ is central, i.e., iff q is scalar (class (a)) or pure (class (b): if $q = v$ is pure then $\bar{v} = -v$ and the square of the map is conjugation by $v \bar{v}^{-1} = -1$, trivial). For pure p , $v \bar{p} v^{-1} = -v p v^{-1} = p$ iff p anticommutes with v , i.e., $p \perp v$; this identifies the fixed spaces. Conjugacy of the class-(b) involutions follows from the transitivity of the rotation action on pure directions. For class (a): the condition on σ says σ equals the reduced word for σ^{-1} ; writing $\sigma = v c \rho(v^*)$ as in Lemma 2.2 (with all four letters now inverted in the mirror) forces the middle letter to satisfy $c = c^{-1}$ formally, which no letter does, unless one allows the empty middle—i.e., $\sigma = v \cdot \rho(v^*)$, whose evaluation is $h(v) \overline{h(v)} = N(h(v))$, a scalar, and whose “identity” is the trivial $N^2 = N^2$. $\square$

5.5 A remark on the null conic

Setting $x^2 + y^2 + z^2 = 0$ defines a smooth conic $\cong \mathbb{P}^1$ over $\mathbb{C}$, on which each triple $(f_1 : f_j : f_k)$, for fixed (t, u) , restricts to a self-map of the conic (the identity (3) says exactly that the image lies on the conic). These self-maps are equivariant for the torus of §5 acting through rotation about the i -axis, and equivariant self-maps of $\mathbb{P}^1$ are severely constrained (monomial in suitable coordinates). We have not pursued this, but note it as a possible independent route to the injectivity and census results of §6: distinct words should give distinct degrees-with-twists on the conic.

6 The census

6.1 Injectivity

Theorem 6.1. *The map h is injective on reduced words in W . Moreover, if $h(\sigma) = \lambda h(\sigma')$ for reduced words σ, σ' and a scalar $\lambda \in \mathbb{R}^\times$, then $\sigma = \sigma'$ and $\lambda = 1$.*

Proof. Suppose $h(\sigma) = \lambda h(\sigma')$. Comparing norms via (4), σ and σ' have the same type and $\lambda^2 = 1$, so $\lambda = \pm 1$. Distinct reduced words represent distinct elements of F_2 . Specialize

$$(t, u, x, y, z) = (\sqrt{2}, 1, \sqrt{2}, 1, 0),$$

so that $h(\alpha) = \sqrt{2}+i$ and $h(\beta) = \sqrt{2}+j$, each of norm 3. The conjugation action of $\sqrt{2}+i$ on pure quaternions is the rotation about the i -axis through the angle ψ with $\cos \psi = 2 \cdot \frac{2}{3} - 1 = \frac{1}{3}$; likewise $\sqrt{2}+j$ rotates about the j -axis through $\arccos \frac{1}{3}$. By Świerczkowski's theorem [11]—two rotations through the angle $\arccos \frac{1}{3}$ (the dihedral angle of the regular tetrahedron) about perpendicular axes generate a free group—these two rotations generate a free group; hence the induced homomorphism $F_2 \rightarrow \text{SO}(3)$ is injective, and distinct reduced words have distinct images. But real-proportional quaternions induce the *same* rotation, so the specialized values of $h(\sigma)$ and $h(\sigma')$, being proportional, correspond to the same group element, forcing $\sigma = \sigma'$ in F_2 and hence as reduced strings. Finally $h(\sigma) = -h(\sigma)$ is impossible for a nonzero integral polynomial, so $\lambda = 1$. $\square$

Remark 6.2 (A simplification of Świerczkowski's proof). Świerczkowski's argument begins by conjugating the word so that it starts with a rotation about the first axis. This normalization can be dispensed with: tracking instead a starting row *adapted to the first letter* of the word (e_2 for the first generator, which moves the 2, 3-plane; e_1 for the second), the 3-scaled matrix product preserves rows of the shape $(d_1, d_2, \sqrt{2}d_3)$ with integer d_i , and a single four-case congruence invariant modulo 3—carrying $3 \nmid d_3$, divisibility by 3 of the inert coordinate, and one companion congruence—propagates along every reduced word in one left-to-right pass. We have formally verified the theorem in this form (Remark 7.4).

6.2 Identities up to sign

Two triples of polynomials determine “the same” identity when they determine the same multiset of three squares. As a first, coarser bookkeeping, call triples *sign-equivalent* when they differ by a global sign and/or by the simultaneous negation of the last two entries:

$$(f_1, f_j, f_k) \sim \pm(f_1, f_j, f_k), \pm(f_1, -f_j, -f_k).$$

These are exactly the changes effected by the quaternion symmetries $p \mapsto \pm p$, $p \mapsto \pm ipi^{-1}$, which visibly preserve the construction; the finer sign changes and the transposition of f_j with f_k are treated in §6.3.

Theorem 6.3. *Sign-equivalence classes of identities of type (m, n) are in bijection with orbits of ι_β on the type- (m, n) part of S ; each orbit has size 2, and the number of classes is*

$$c(m, n) = R\left(m, \left\lceil \frac{n}{2} \right\rceil - 1\right), \quad \text{where} \quad \sum_{m, k \geq 0} R(m, k) s^m q^k = \frac{1 + s}{1 - s - q - 3sq}.$$

Proof. Bijection. For $\sigma \in S$, conjugation by i gives $h(\iota_\beta \sigma) = ih(\sigma)i^{-1} = (f_1, -f_j, -f_k)$ (proof of Proposition 4.1), so ι_β -orbits map to sign-equivalence classes. Conversely, if the triples of σ, σ' are sign-equivalent then $h(\sigma') = \pm h(\sigma)$ or $\pm ih(\sigma)i^{-1} = \pm h(\iota_\beta \sigma)$, and Theorem 6.1 forces $\sigma' \in \{\sigma, \iota_\beta \sigma\}$. Each orbit has size exactly 2: $\iota_\beta \sigma = \sigma$ would require σ free of β -letters, impossible since the middle block of Lemma 2.2 is a β -power.

Count. By Lemma 2.2, elements of S of type (m, n) correspond to pairs (v, ε) : parity forces $|\varepsilon| \in \{1, 2\}$ with $|\varepsilon| \equiv n \pmod{2}$, the sign of ε is free, and v ranges over reduced words with m letters from $\{\alpha, \alpha^{-1}\}$ and $k = (n - |\varepsilon|)/2$ letters from $\{\beta, \beta^{-1}\}$, not ending in $\beta^{-\text{sgn } \varepsilon}$. The two sign choices give equinumerous sets (exchanged by ι_β applied to v), so $|S(m, n)| = 2R(m, k)$

with $R(m, k)$ the number of reduced words with m α -letters and k β -letters not ending in β^{-1} , and $k = \lceil n/2 \rceil - 1$. Dividing by the orbit size 2 gives $c(m, n) = R(m, k)$.

Generating function. Grade letters $\alpha^{\pm 1}$ by s and $\beta^{\pm 1}$ by q . Let g (resp. h') be the generating function for nonempty reduced words ending in a fixed one of α, α^{-1} (resp. β, β^{-1}); by the symmetry inverting all letters these depend only on the letter class. Appending a final letter to any reduced word not ending in that letter's inverse,

$$g = s(1 + g + 2h'), \quad h' = q(1 + 2g + h').$$

Solving, $h' = q(1 + s)/(1 - s - q - 3sq)$, and the reduced words not ending in β^{-1} are counted by $1 + 2g + h'$, which simplifies to $(1 + s)/(1 - s - q - 3sq)$. $\square$

Corollary 6.4 (Growth). (i) $c(m, 2k+1) = c(m, 2k+2)$ for all $k \geq 0$: the counts advance in equal steps of two in n .

(ii) $R(1, k) = 4k + 2$; hence $c(1, n) = 2n$ for n odd and $2n - 2$ for n even.

(iii) For fixed m , extracting $[q^k]$ first gives $R(m, k) = [s^m](1 + s)(1 + 3s)^k(1 - s)^{-(k+1)}$, a polynomial in k of degree m with leading coefficient $4^m/m!$; thus $c(m, n)$ grows like $\frac{2^m}{m!}n^m$ for fixed m . For example $R(2, k) = 8k^2 + 4k + 2$.

(iv) Along the diagonal, $R(k, k)^{1/k} \rightarrow \min_{0 < s < 1} \frac{1+3s}{s(1-s)} = 9$ (attained at $s = \frac{1}{3}$); the count is exponential only when m grows in proportion to n .

Proof. (i) is the ceiling in Theorem 6.3. (ii), (iii): expand $(1 + s)(1 + 3s)^k(1 - s)^{-(k+1)}$; the coefficient of s^m is $\sum_{a+b \leq m}$ of products of binomials $\binom{k}{a} 3^a \binom{k+b}{b}$ (with the $(1 + s)$ shift), a polynomial in k of degree m whose top term is $k^m \sum_{a+b=m} \frac{3^a}{a!b!} = \frac{(3+1)^m}{m!} k^m$. (iv) is the standard exponential-rate computation for $[s^k]H(s)^k$ with $H(s) = (1 + 3s)/(1 - s)$: the rate is $\inf_{0 < s < 1} H(s)/s$, and calculus locates the infimum at $s = \frac{1}{3}$ with value 9. $\square$

$c(m, n)$	$n=1$	2	3	4	5	6	7	8
$m=0$	1	1	1	1	1	1	1	1
1	2	2	6	6	10	10	14	14
2	2	2	14	14	42	42	86	86
3	2	2	22	22	106	106	318	318
4	2	2	30	30	202	202	838	838

For fixed m the counts grow polynomially in n (linearly for $m = 1$), while the diagonal entries $1, 6, 42, 318, \dots$ grow exponentially; growth in $m + n$ is exponential only along diagonals.

6.3 Fine equivalence

The finest natural equivalence identifies triples with the same multiset $\{f_1^2, f_j^2, f_k^2\}$, i.e., allows independent sign changes and permutations. By Proposition 4.1, f_1 has even and f_j, f_k odd (y, z) -parity, so (nondegeneracy granted) a permutation can only transpose f_j with f_k . The independent sign changes and the transposition are effected by conjugating $h(\sigma)$ by units in $\{1, i, j, k\}$ and by $\frac{j+k}{\sqrt{2}}$; each such conjugation carries the letter values to letter values of a

substituted variable system (for instance conjugation by j fixes $x + yj + zk$ only after $z \mapsto -z$), and the arguments of Theorem 6.3 then localize any possible coincidence:

Proposition 6.5. *Under the finest equivalence, distinct sign-equivalence classes of type (m, n) can coincide only in pairs $\{\sigma\}, \{\iota_\alpha \sigma\}$, and only if σ contains equally many β and β^{-1} letters; in particular n must be even. Hence for n odd the fine count equals $c(m, n)$, and for n even it lies between $\frac{1}{2}c(m, n)$ and $c(m, n)$.*

Proof. Suppose the triples of $\sigma, \sigma' \in S$ agree up to signs and the transposition. As noted, the sign patterns and the transposition are realized by $p \mapsto \pm gpg^{-1}$ with $g \in \{1, i, j, k, \frac{j+k}{\sqrt{2}}, \dots\}$; composing with the analysis already done for $g \in \{1, i\}$, the remaining cases give an equation of the form

$$h(\sigma') = \pm h(\iota_\alpha \sigma)|_{\text{sub}}, \quad (7)$$

where “sub” is one of the substitutions $z \mapsto -z$, $y \mapsto -y$, $y \leftrightarrow z$, or $(y, z) \mapsto (z, -y)$ (compositions of the reflection and rotation of the w -plane induced by the conjugations). Specializing (7) at a point fixed by the substitution and at which the pair of rotations is free— $(\sqrt{2}, 1, \sqrt{2}, 1, 0)$ for the substitutions fixing $z = 0$, and $(\sqrt{2}, 1, 2, 1, 1)$ (norms 3 and 6, angles again $\arccos \frac{1}{3}$, axes i and the perpendicular $(0, 1, 1)$; Świerczkowski’s theorem, being stated for arbitrary perpendicular axes, applies directly) for $y \leftrightarrow z$ —forces, as in Theorem 6.1, the word equation $\sigma' = \iota_\alpha \sigma$, whereupon (7) becomes a symmetry requirement on the single polynomial quaternion $h(\sigma')$.

To constrain that requirement, set $u = 0$. Each α -letter evaluates to the central element t , and the β -letters evaluate to the commuting elements $x \pm wj$ (they commute since both lie in $\mathbb{R}[x, wj]$ with $(wj)^2 = -w\bar{w}$ central); hence

$$h(\tau)|_{u=0} = t^{2m} (x + wj)^p (x - wj)^q, \quad p + q = n,$$

where p, q count the β, β^{-1} letters of τ . If $p \neq q$, comparing the coefficients of x^{n-1} on the two sides of (7) at $u = 0$ gives $(p - q)w = \pm(p - q)w'$, with w' the image of w under the substitution $(\bar{w}, -\bar{w}, i\bar{w}, \text{ or } -i\bar{w})$; since w is not a scalar multiple of $\bar{w}$, and $w = \pm i\bar{w}$ is impossible over $\mathbb{R}$, the substitution must be trivial—contradiction. So any residual coincidence requires $p = q$.

(The case $(y, z) \mapsto (z, -y)$, i.e. $w \mapsto -i\bar{w}$: here the same comparison gives $w = \mp i\bar{w}$, impossible outright, so this substitution never occurs, balanced or not.)

For n odd, $p \neq q$ always, so no coincidence beyond sign-equivalence occurs and the fine count is exactly $c(m, n)$. $\square$

Theorem 6.6 (Freeness for the three coordinate axes). *The rotations through $\arccos \frac{1}{3}$ about the three coordinate axes generate a free group of rank three.*

Proof. Scale each rotation by 3, so that the generators and their inverses are the six integer- $\sqrt{2}$ matrices with diagonal block $\begin{pmatrix} 1 & \mp 2\sqrt{2} \\ \pm 2\sqrt{2} & 1 \end{pmatrix}$ in the moving plane and 3 in the fixed slot. Reduce modulo 3, working in $\mathbb{F}_9 = \mathbb{F}_3(\sqrt{2})$ (the prime 3 is inert in $\mathbb{Z}[\sqrt{2}]$, since 2 is not a square modulo 3). Each scaled generator becomes a matrix of rank one: its 2×2 block has determinant $1 + 8 = 9 \equiv 0$, while the block itself is nonzero. A product of rank-one matrices $M_1 \cdots M_n$ equals the outer product of the column of M_1 with the row of M_n , multiplied by the scalars $r_i c_{i+1}$ pairing the row of each factor with the column of the next; it vanishes if and only if some such junction scalar vanishes. For the six generators there are thirty junctions

compatible with reducedness (all ordered pairs except a generator followed by its own inverse), and each scalar is a unit modulo 3: same-generator junctions give $\pm(\text{trace}) = \mp 1$, and cross junctions pair the two $\pm 2\sqrt{2}$ entries meeting in the shared coordinate, giving $\pm 8 \equiv \pm 1$ — a finite verification. Hence for every nonempty reduced word the scaled product is nonzero modulo 3, while the scaled identity $3^n \cdot I$ vanishes modulo 3 for $n \geq 1$; so no nonempty reduced word represents the identity rotation. $\square$

Remark 6.7. The proof refines the odd-modulus half of Świerczkowski’s method: his tracking of d_2 modulo the odd part of b [12] is, in coordinates, the statement that the scaled generators are rank one there and that consecutive factors mesh. The refinement is the observation that the meshing is a purely local, finite condition, indifferent to the number of coordinate-axis generators; the same verification yields freeness for the rotations about all three axes, and more generally at any angle $\arccos \frac{a}{b}$ (in lowest terms) whose denominator has an odd prime factor. The dyadic angles $b = 2^m$, where his argument requires the delicate valuation bookkeeping, are not covered and are not needed here.

Corollary 6.8. *Under the finest equivalence, no two distinct sign-equivalence classes of type (m, n) are identified by any of the sign patterns; identification could occur only through the transposition of f_j with f_k , only between the classes of σ and $\iota_\alpha \sigma$ with σ balanced, and only if $f_j(\sigma)$ is symmetric or antisymmetric under the interchange of y and z .*

Proof. By the proof of Proposition 6.5 the sign-pattern cases reduce to a symmetry requirement $h(\sigma') = \pm h(\sigma')|_{z \mapsto -z}$ (or $y \mapsto -y$) on a single word. Specialize at $(\sqrt{2}, 1, 2, 1, 1)$: the substitution carries the second generator to the rotation about the axis $(0, 1, -1)$, and the three axes $e_1, (0, 1, 1), (0, 1, -1)$ are mutually orthogonal, so by Theorem 6.6 (transported by a frame rotation) the three rotations through $\arccos \frac{1}{3}$ about them generate a free group Γ_3 on generators $A, B, \tilde{B}$. The specialized requirement equates a reduced word over $\{A^{\pm 1}, B^{\pm 1}\}$ with a reduced word over $\{A^{\pm 1}, \tilde{B}^{\pm 1}\}$; in a free group, equal reduced words are letterwise identical, so both words use the letters $A^{\pm 1}$ only. But every word of S contains β -letters (its middle block), a contradiction. The transposition case survives this argument only through the substitution $y \leftrightarrow z$, whose residual condition, by the relation $f_k(\sigma) = f_j(\sigma)|_{y \leftrightarrow z, w \mapsto -w}$, amounts to the stated symmetry of $f_j(\sigma)$. $\square$

Conjecture 6.9. *No word of S has f_j symmetric or antisymmetric under the interchange of y and z ; consequently the fine count equals $c(m, n)$ for all (m, n) . We have verified this by machine for all words through length 12.*

Remark 6.10. The fine-equivalence question has an exact integer-level antecedent: Pall [5, Lemma 20, Theorem 8] determines when a quaternion obtained from a pure x by sign-changes and interchanges of coordinates is reachable as $tx\bar{t}/N(t)$, with explicit binary quadratic forms effecting the coordinate permutations. The polynomial question treated here is the word-level refinement of his.

7 Generalizations, specializations, and questions

7.1 More letters

Let $W_{r,s}$ be the free monoid on letters $\alpha_1^{\pm 1}, \dots, \alpha_r^{\pm 1}, \beta_1^{\pm 1}, \dots, \beta_s^{\pm 1}$, evaluated by

$$h(\alpha_a^{\pm 1}) = t_a \pm iu_a, \quad h(\beta_b^{\pm 1}) = x_b \pm (y_b j + z_b k),$$

and let $S_{r,s}$ be the reduced words fixed by reversal composed with inversion of all α -letters. Lemma 3.1 and Theorem 3.2 apply verbatim— θ fixes every $h(\beta_b^{\pm 1})$ and exchanges each $h(\alpha_a)$ with $h(\alpha_a^{-1})$ —so each $\sigma \in S_{r,s}$ yields

$$\prod_{a=1}^r (t_a^2 + u_a^2)^{2m_a} \prod_{b=1}^s (x_b^2 + y_b^2 + z_b^2)^{n_b} = f_1^2 + f_j^2 + f_k^2,$$

with every α_a -exponent even (the letters α_a pair off across the middle as in Lemma 2.2, whose proof is unchanged: the middle block is a power of a single β_b). Theorem 6.1 extends by choosing rotations about axes in general position (freeness for generic tuples reduces to the two-generator case by specialization); the census becomes a transfer-matrix computation on $2r + 2s$ letters which we do not carry out here.

7.2 Clifford algebras

The mechanism has an evident Clifford-theoretic template: in the Clifford algebra of a quadratic space, the transpose anti-involution has a large fixed space, the norm is multiplicative on the Clifford group, and conjugation implements $O(k)$ on vectors. Words in suitable generators, symmetrized as here, should produce sum-of- N -squares identities for products of quadratic forms with N the dimension of an appropriate fixed space. Whether anything arises beyond restatements of the spin representation theory—in particular, whether any *census* phenomenon of the present kind persists—we leave open.

7.3 Integer specializations

Specializing (3) at integers gives constructive access to representations: whenever $c = t^2 + u^2$ and $N = x^2 + y^2 + z^2$, the integer $c^{2m} N^n$ is represented by the specialized triple, compatibly with the classical theorem of Legendre and Gauss. We caution against expecting more, and can now say precisely why: by Pall's theorem [5, §8], every representation of hm^2 satisfying his local condition already has the form $tx\bar{t}$, so at the level of integers the family produces nothing outside the classical quaternionic parametrization; and projectively the polynomial maps of the family scale the ternary form by $(t^2 + u^2)^{2m}$, landing in the rational orthogonal group of $x^2 + y^2 + z^2$ —parametrized by quaternions in Pall's companion work on the rational automorphs of $x_1^2 + x_2^2 + x_3^2$ (announced in [5])—where questions of transitivity on representations are the province of spinor genus theory and are settled by other means. The one Diophantine remark the family does make cleanly: the rotations reached by words are exactly those rational rotations whose denominators are norms arising from the specialized letters—a parametrized subgroup, not the full rational orthogonal group.

7.4 Questions

Question 7.1. Does any word of S have f_j symmetric or antisymmetric under the interchange of y and z (Conjecture 6.9)? Equivalently: does any identity of the family contain a repeated square, up to the symmetries of the construction?

Question 7.2. Does the restriction-to-the-null-conic map of §5 separate words, giving a census proof independent of free rotation groups?

Question 7.3. In the Clifford setting, for which quadratic spaces does an exact word census survive?

Remark 7.4 (Verification). All identities displayed or tabulated in this paper were generated symbolically from their words and verified against Theorem 3.2 and (4) by machine; the census values through length 12 were computed by exhaustive enumeration and canonicalization under the fine equivalence. Beyond this, the mathematical core of the paper has been formally verified in Lean 4 against Mathlib, over an arbitrary commutative ring where so stated: Theorem 3.2 (by the second proof) and Corollary 3.5; the evenness of the α -count; Proposition 4.1; the $[2, 3, 3]$ obstruction of the remark following Proposition 4.2; Świerczkowski’s freeness theorem for the $\arccos \frac{1}{3}$ pair, in the simplified form of Remark 6.2; Theorem 6.1, at the level of the generic polynomial evaluation and up to sign; Theorem 6.3 in full—the classification of identities up to sign-equivalence, the bisection, the identity $|S(m, n)| = 2R(m, \lceil n/2 \rceil - 1)$ with the recurrence for R and its closed forms for $m \leq 2$, the count of classes as a single statement, and kernel-checked enumeration instances through length 7; and Theorem 6.6, by the mesh argument exactly as printed, from which Świerczkowski’s theorem for the pair is re-derived as a corollary, so that the freeness of the pair carries two independent machine-checked proofs. The formalization comprises ninety-one theorems in four files, with no unverified assumptions beyond the standard axioms of Mathlib; it, the paper source, and all scripts are available at github.com/DavidVFeldman/three-squares-repo and archived at DOI:10.5281/zenodo.21943651.

A Tables of identities in complex coordinates

Recall the dictionary: $\zeta = t + iu$, $w = y + iz$, quaternions are pairs $z_1 + z_2j$ with the multiplication (5), and the letters evaluate to

$$h(\alpha^{\pm 1}) = \zeta^{\pm} \text{ (i.e. } \zeta \text{ or } \bar{\zeta}), \quad h(\beta^{\pm 1}) = x \pm wj.$$

For $\sigma \in S$ of type (m, n) we tabulate $f_1 = z_1$ (a real polynomial: invariant under $\zeta \leftrightarrow \bar{\zeta}$, $w \leftrightarrow \bar{w}$) and $F = z_2 = f_j + if_k$; the identity reads

$$(\zeta \bar{\zeta})^{2m} (x^2 + w\bar{w})^n = f_1^2 + F\bar{F}.$$

To recover the real form, substitute $\zeta = t + iu$, $w = y + iz$ and take $f_j = \operatorname{Re} F$, $f_k = \operatorname{Im} F$.

Each table lists one representative per mirror pair: the involution ι_α (equivalently, the substitution $u \mapsto -u$, i.e. $\zeta \leftrightarrow \bar{\zeta}$ in the formulas) carries each row to a distinct identity of the same type, its *mirror*, which we do not print. The involution ι_β (the substitution $w \mapsto -w$ together with $F \mapsto -F$... more precisely $(f_1, f_j, f_k) \mapsto (f_1, -f_j, -f_k)$) gives the

sign-equivalent partner and is likewise suppressed. The $m = 0$ family β^n is described in closed form in §5 and omitted. Note in every row the weight-one structure of F (each monomial has w -degree minus $\bar{w}$ -degree equal to 1) promised in §5.

Type $(m, n) = (1, 1)$ (all 2 classes: the row below and its mirror)

σ	f_1	F
$\alpha^{-1}\beta\alpha$	$\zeta\zeta x$	$\zeta^2 w$

Type $(m, n) = (1, 2)$ (all 2 classes: the row below and its mirror)

σ	f_1	F
$\alpha^{-1}\beta^2\alpha$	$\zeta\zeta(-\bar{w}w + x^2)$	$2\zeta^2 wx$

Type $(m, n) = (1, 3)$ (all 6 classes: the 3 rows below and their mirrors)

σ	f_1	F
$\alpha^{-1}\beta^3\alpha$	$\zeta\zeta x(-3\bar{w}w + x^2)$	$\zeta^2 w(-\bar{w}w + 3x^2)$
$\beta\alpha^{-1}\beta\alpha\beta$	$x(-\bar{\zeta}^2\bar{w}w - \bar{\zeta}\bar{w}\zeta w + \bar{\zeta}\zeta x^2 - \bar{w}\zeta^2 w)$	$w(\bar{\zeta}^2 x^2 + 2\bar{\zeta}\zeta x^2 - \bar{w}\zeta^2 w)$
$\beta^{-1}\alpha^{-1}\beta\alpha\beta^{-1}$	$x(\bar{\zeta}^2\bar{w}w - \bar{\zeta}\bar{w}\zeta w + \bar{\zeta}\zeta x^2 + \bar{w}\zeta^2 w)$	$w(\bar{\zeta}^2 x^2 - 2\bar{\zeta}\zeta x^2 - \bar{w}\zeta^2 w)$

Type $(m, n) = (1, 4)$ (all 6 classes: the 3 rows below and their mirrors)

σ	f_1	F
$\alpha^{-1}\beta^4\alpha$	$\zeta\zeta(\bar{w}^2 w^2 - 6\bar{w}wx^2 + x^4)$	$4\zeta^2 wx(-\bar{w}w + x^2)$
$\beta\alpha^{-1}\beta^2\alpha\beta$	$-2\bar{\zeta}^2\bar{w}wx^2 + \bar{\zeta}\bar{w}^2\zeta w^2 - 2\bar{\zeta}\bar{w}\zeta wx^2 + \bar{\zeta}\zeta x^4 - 2\bar{w}\zeta^2 wx^2$	$2wx(\bar{\zeta} + \zeta)(\bar{\zeta}x^2 - \bar{w}\zeta w)$
$\beta^{-1}\alpha^{-1}\beta^2\alpha\beta^{-1}$	$2\bar{\zeta}^2\bar{w}wx^2 + \bar{\zeta}\bar{w}^2\zeta w^2 - 2\bar{\zeta}\bar{w}\zeta wx^2 + \bar{\zeta}\zeta x^4 + 2\bar{w}\zeta^2 wx^2$	$2wx(\bar{\zeta} - \zeta)(\bar{\zeta}x^2 + \bar{w}\zeta w)$

Type $(m, n) = (2, 1)$ (all 2 classes: the row below and its mirror)

σ	f_1	F
$\alpha^{-2}\beta\alpha^2$	$\zeta^2\zeta^2 x$	$\zeta^4 w$

Type $(m, n) = (2, 2)$ (all 2 classes: the row below and its mirror)

σ	f_1	F
$\alpha^{-2}\beta^2\alpha^2$	$\zeta^2\zeta^2(-\bar{w}w + x^2)$	$2\zeta^4 wx$

Type $(m, n) = (2, 3)$ (14 classes in all; 3 sample rows)

σ	f_1	F
$\alpha^{-2}\beta^3\alpha^2$	$\zeta^2\zeta^2 x(-3\bar{w}w + x^2)$	$\zeta^4 w(-\bar{w}w + 3x^2)$
$\alpha^{-1}\beta\alpha^{-1}\beta\alpha\beta\alpha$	$\bar{\zeta}\zeta x(-\bar{\zeta}^2\bar{w}w - \bar{\zeta}\bar{w}\zeta w + \bar{\zeta}\zeta x^2 - \bar{w}\zeta^2 w)$	$\bar{\zeta}^2 w(\bar{\zeta}^2 x^2 + 2\bar{\zeta}\zeta x^2 - \bar{w}\zeta^2 w)$
$\alpha^{-1}\beta\alpha\beta\alpha^{-1}\beta\alpha$	$\bar{\zeta}\zeta x(-\bar{\zeta}^2\bar{w}w - \bar{\zeta}\bar{w}\zeta w + \bar{\zeta}\zeta x^2 - \bar{w}\zeta^2 w)$	$\bar{\zeta}^2 w(-\bar{\zeta}^2\bar{w}w + 2\bar{\zeta}\zeta x^2 + \zeta^2 x^2)$

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